Veer Pratap Singh

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SUMMARY

Full-stack developer who builds AI into real products, from React interfaces and Node.js APIs to machine learning and generative AI features people can trust. Enjoys owning a feature end to end, questioning data quality, and explaining technical decisions clearly to any audience. Always learning, currently exploring agentic AI.

EDUCATION

Vellore Institute of Technology, Amaravati , Andhra Pradesh 2023 – 2027
Bachelor of Technology in Computer Science Engineering

Nalanda Academy, Kota, Rajasthan 2021 – 2022
Class XII: 84.8%

Nalanda Academy, Kota, Rajasthan 2019 – 2020
Class X: 86.8%

TECHNICAL SKILLS

Programming Languages: Python, SQL, Java, JavaScript, Bash
Generative & Agentic AI: Generative AI APIs, Prompt Engineering, Agentic AI, Multi-Agent Systems, OCR
Machine Learning & NLP: PyTorch, Scikit-learn, Deep learning, Transfer Learning, Model Evaluation, NLP
Data Analysis & Visualization: NumPy, Pandas, Matplotlib, Seaborn, EDA, Data Preprocessing, Streamlit
Cloud & Tools: AWS, Git, GitHub Actions (CI/CD), Docker, Linux, Flask, REST APIs
Databases: MySQL, PostgreSQL, MongoDB

EXPERIENCE

Software Developer Intern Jun 2025 – Dec 2025
Cipherium Private Limited *Remote*

- Developed and deployed web applications in Python and JavaScript from design through production, using systematic debugging to diagnose and resolve production issues.
- Designed CI/CD pipelines for automated backend deployment on Linux servers, cutting deployment time by 30%.
- Overhauled documentation and code structure, cutting new-developer onboarding time by 25%.

PROJECTS

Klaro | Generative AI API, React, Node.js, Express, Multer Live Demo

- Built a generative AI document summarizer: uploaded PDFs and scanned images are routed by file type to pdf-parse or Tesseract.js OCR for text extraction, then passed to a Generative AI API.
- Engineered prompts around the user's chosen summary length to return structured, actionable summaries through a React drag-and-drop frontend and a Node.js/Express backend.

Agentic Security Operations Center (SOC) | Agentic AI, Multi-Agent Systems In Progress

- Designing a multi-agent SOC in which cooperating AI agents work together on security operations, scoped as a research-level system targeting an international conference paper.
- Surveying existing agentic-security research to differentiate the design, select models, and plan the implementation roadmap.

PulmoSight | Python, PyTorch, Transfer Learning, CNN, Flask, Grad-CAM GitHub

- Built an end-to-end tuberculosis screening pipeline: prepared 16,800+ Indian chest X-rays
- Validated on held-out data and added Grad-CAM heatmaps with a decision threshold in a Flask app so every prediction is explainable, supporting transparent and responsible AI.

Analytica Studio | Python, Numpy, Streamlit, Pandas, Scikit-learn Live Demo

- Built a Streamlit platform that automates exploratory data analysis (EDA) and data preprocessing with Pandas and Scikit-learn, cutting manual analysis time by 60% on datasets with 100K+ rows.

CERTIFICATIONS & LANGUAGES

Certifications: AWS Cloud Architecture | AWS Cloud Foundations | Full Stack Dev
Languages: Hindi (Native), English (Professional), German (Beginner)

LEADERSHIP & ACHIEVEMENTS

- **Hackathon:** First Runner-up, National-Level Hackathon at Woxsen University.
- **Leadership:** President of Team Next Nexus (organized 50+ technical events); Core Team Member of VLaunchpad and VTOPIA.
- **Workshops:** Conducted technical workshops on Firebase and Introduction to IoT for 100+ students.
- **Organizer:** Key organizer for TEDxVITAP; active member of the University Social Media Committee.